

PORTLAND CEMENT CEM I 52.5 N		
DESCRIPTION	LIMITATIONS BS EN 197 - 1 (2011) and ES 4756 - 1 (2022)	OBTAINED RESULTS
PHYSICAL TEST		
Soundness (Expansion) (mm)	≤ 10 MAX.	1.0
Fineness , Blaine (cm ² /g)	..	3650
Insoluble Residue (%)	≤ 5.0	3.2
Initial setting time (minute)	≥ 45 MIN.	175
Water consistency (%)	..	26
Compressive Strength (Mpa)	2 days	≥ 20 MIN. 22.5
	28 days	≥ 52.5 MIN. 55.0
CHEMICAL ANALYSIS		
Silicone Dioxide (SiO2) %	..	19.77
Aluminium Oxide (Al2O3) %	..	5.01
Ferric Oxide (Fe2O3) %	..	3.44
Calcium Oxide (CaO) %	..	61.78
Magnesium Oxide (MgO) %	..	1.15
Sulfur Trioxide (SO3) %	≤ 4.0	3.55
Sodium Oxide (Na2O) %	..	0.37
Potassium Oxide (K2O) %	..	0.23
Chloride (Cl) %	≤ 0.1	0.05
Loss On Ignition (LOI) %	≤ 5.0	3.83

PORTLAND CEMENT

CEM I 42.5N

DESCRIPTION	LIMITATIONS BS EN 197 - 1 (2011) and ES 4756 - 1 (2022)	OBTAINED RESULTS
-------------	---	------------------

PHYSICAL TEST

Soundness (Expansion) (mm)	≤ 10	0.0
Fineness , Blaine (cm ² /g)	..	3600.0
Insoluble Residue (%)	≤ 5.0	3.2
Initial setting time (minute)	≥ 60	165.0
Water consistency (%)	..	28.1
Compressive Strength (Mpa)	2 days	≥ 10
	28 days	≥ 42.5
		19.0
		48.0

CHEMICAL ANALYSIS

Silicone Dioxide (SiO ₂) %	..	19.50
Aluminium Oxide (Al ₂ O ₃) %	..	4.89
Ferric Oxide (Fe ₂ O ₃) %	..	3.64
Calcium Oxide (CaO) %	..	36.10
Magnesium Oxide (MgO) %	..	1.15
Sulfur Trioxide (SO ₃) %	≤ 3.5	3.30
Sodium Oxide (Na ₂ O) %	..	0.33
Potassium Oxide (K ₂ O) %	..	0.24
Chloride (Cl) %	≤ 0.1	0.05
Loss On Ignition	≤ 5	3.10
Hexavalent Chromium (PPM)	≤ 2	1.80

PORTLAND CEMENT
CEM II A-P 32.5N

DESCRIPTION		LIMITATIONS BS EN 197 - 1 (2011) and ES 4756 - 1 (2022)	OBTAINED RESULTS
PHYSICAL TEST			
Soundness (Expansion) (mm)		≤ 10	1.0
Fineness , Blaine (cm ² /g)		..	3950
Initial setting time (minute)		≥ 75	165
Water consistency (%)		..	28
Compressive Strength (Mpa)	2 days	≥ 16	17.0
	28 days	≥ 32.5	37.0
CHEMICAL ANALYSIS			
Silicone Dioxide (SiO ₂) %		..	21.19
Aluminium Oxide (Al ₂ O ₃) %		..	6.15
Ferric Oxide (Fe ₂ O ₃) %		..	6.16
Calcium Oxide (CaO) %		..	54.45
Magnesium Oxide (MgO) %		..	1.68
Sulfer Trioxide (SO ₃) %		≤ 3.5	3.20
Sodium Oxide (Na ₂ O) %		..	0.85
Potassium Oxide (K ₂ O) %		..	0.27
Chloride (Cl) %		≤ 0.1	0.03

CLINKER SPECIFICATION				
	MIN	MAX	TYPICAL	RATIO
C3S	53	57	55.00	%
C2S	17	21	19.00	%
C4AF	10.5	12.5	11.50	%
C3A	7.5	8.5	8.00	%
F.L	1.8	2.2	2.00	%
LOI	0.2	0.4	0.30	%
IR	0.1	0.3	0.20	%
MgO	0.8	1.2	1.00	%
SO3	1.3	1.6	1.45	%
CL	0.01	0.04	0.03	%
Alkalis Equ.	0.55	0.65	0.60	%